STOCHASTIC METHODS IN WATER RESOURCES

Unit 2: Hydrological statistics and extremes
Lecture 7: Probability distributions of extremes, distribution selection

Luis Alejandro Morales, Ph.D.

Universidad Nacional de Colombia
Department of Civil and Agriculture Engineering

September 7, 2025

Gumbel distribution

The **Gumbel distribution** is equivalent to the to asymptotic distribution type I and result from a basic exponential distribution. The Gumbel distribution has two parameters: the scale parameter α and the position parameter b, and is given for maximum values represented by X, where:

$$f_X(x) = \frac{1}{\alpha} e^{\left[-\frac{x-b}{\alpha} - e^{\left(-\frac{x-b}{\alpha}\right)}\right]}, \text{ for } -\infty < x < \infty$$

The cdf is:

$$F_X(x) = e^{\left[-e^{\left(-\frac{x-b}{\alpha}\right)}\right]}$$

and $\mu_X = b + 0.5772\alpha$ and $\sigma_X^2 = \frac{\pi^2\alpha^2}{6}$ and the asymmetry coefficient $\gamma_1 = 1.1396$. The quantile of this distribution is:

$$x(F) = b - \alpha \ln(-\ln F)$$

where $x(F)$ is the value of X for which the cdf value is F. This equation is equivalent to:

$$x_T = b - \alpha \ln\left[-\ln\left(1 - \frac{1}{T}\right)\right]$$

where T is the return period and x_T is the magnitude of the extreme event associated to this.

Gumbel distribution

The parameter estimation in the Gumbel distribution can be performed using different methods such as the Moments and the Maximum-Likelihood. The frequency factor K_T for this distribution when $n \to \infty$, converge asyntotically to:

$$K_T = -\frac{\sqrt{6}}{\pi}\left[0.5772 + \ln\left(\ln\left(1 - \frac{1}{T}\right)\right)\right]$$

On the other hand, for a value of the annual series, the frequency factor as a function of the size series, can be estimated as:

$$K_m = \frac{y_m - \bar{y}}{s_y}$$

where:

$$y_m = -\ln\left[-\ln\left(\frac{n+1-m}{n+1}\right)\right]$$

Note that the data are sorted in descending order; from the largest $(m = 1)$ to the smallest $(m = n)$. $\bar{y} = \frac{1}{n}\sum_{m=1}^{n} y_m$ and $s_y^2 = \frac{1}{n}\sum_{m=1}^{n}(y_m - \bar{y})^2$. For a return period T, the frequency factor is:

$$K_T = \frac{y_T - \bar{y}}{s_y}$$

where $y_T = -\ln\left[-\ln\left(1 - \frac{1}{T}\right)\right]$. Comparing the two equations to compute K_T the later equations provide larger results of K_T than the former one. This value is progressively larger as much as n decrease. This means that the shorter the series the larger the values of x_T.

Gumbel distribution

Estimation of the standard deviation s_T using different methods:

- Method of Moments

$$s_T^2 = \frac{s_X^2}{n}\left[1 + K_T\gamma_1 + \frac{K_T^2}{4}(\gamma_2 - 1)\right]$$

where $\gamma_1 = 1.1396$ and $\gamma_2 = 5.4002$

- Method of Maximum-Likelihood

$$s_T^2 = \frac{\hat{\alpha}^2}{n}\left[1 + \frac{6}{\pi^2}(1 - 0.5772 + y_T)^2\right]$$

- Method of MPP

$$s_T^2 = \frac{\hat{\alpha}^2}{n}\left[1.1128 + 0.4574 y_T + 0.8046 y_T^2\right]$$

Gumbel distribution

So far, we have analysed annual series of maximum values using the Gumbel distribution. However, this distribution can be used to analyzed minimum values Z so the pdf is:

$$f_Z(z) = \frac{1}{\alpha} e^{\left[\frac{z-b}{\alpha} - e^{\left(\frac{z-b}{\alpha}\right)}\right]}, \text{ for } -\infty < z < \infty$$

and the cdf is:

$$F_Z(z) = 1 - e^{\left[-e^{\left(\frac{x-b}{\alpha}\right)}\right]}$$

The $\mu_Z = b - 0.5772\alpha$, $\sigma_Z^2 = \frac{\pi^2\alpha^2}{6}$ and the asymmetry coefficient is $\gamma_1 = -1.1396$.
The quantile is:

$$z(F) = b + \alpha \ln(-\ln(1 - F))$$

where $z(F)$ is the value of Z for which the cdf value is F. This equation is equivalent to:

$$z_T = b + \alpha \ln\left[-\ln\left(1 - \frac{1}{T}\right)\right]$$

Similarly, the parameter of the distribution can be estimated using different methods.

Gumbel distribution

When $n \to \infty$ asymptotically, the frequency factor K_T converge to :

$$K_T = \frac{\sqrt{6}}{\pi}\left[0.5772 + \ln\left(-\ln\left(1 - \frac{1}{T}\right)\right)\right]$$

On the other hand, for a value of the annual series, the frequency factor as a function of the size series, can be estimated as:

$$K_m = \frac{\bar{y} - y_m}{s_y}$$

where:

$$y_m = \ln\left[-\ln\left(\frac{m}{n+1}\right)\right]$$

Note that the data are sorted in ascending order; from the smallest $(m = 1)$ to the largest $(m = n)$. $\bar{y} = \frac{1}{n}\sum_{m=1}^{n} y_m$ and $s_y^2 = \frac{1}{n}\sum_{m=1}^{n}(y_m - \bar{y})^2$.
For a return period T, the frequency factor is:

$$K_T = \frac{\bar{y} - y_T}{s_y}$$

where $y_T = -\ln\left[-\ln\left(-\frac{1}{T}\right)\right]$.

Pearson type III and Log-Pearson type III distributions

The pdf of the **Pearson type III distribution** is:

$$f_X(x) = \frac{1}{a\Gamma(b)} \left(\frac{x-c}{a}\right)^{b-1} e^{\left[-\left(\frac{x-c}{a}\right)\right]}$$

where the parameters a, b and c are the scale, the shape and the position, respectively. Note that $b > 0$ and that $c < x < \infty$. If $c = 0$, the pdf is reduced to the Gamma distribution. Despite a can be either positive or negative, if $a < 0$ the pdf is bounded above, this means that $x < c$ and it is not suitable to analyse maximum extreme events. Using the methods of Moments:

- Mean

$$\mu_X = c + ba$$

- Variance

$$\sigma_X^2 = ba^2$$

- Asymmetry coefficient

$$\gamma_1 = sgn(a)\frac{2}{\sqrt{b}}$$

  where sgn is the sign function.

Sometimes, the annual series are transformed using the logarithm function and the data is fitted to the Pearson type III distribution, which is equivalent to the **Log-Pearson type III distribution**. This distribution is commonly used to analyse maximum streamflows and requires that $b > 1$ and $a > 0$. The equations to estimate the frequency factor are given in textbook tables.

Weibull distribution

The **Weibull distribution** is equivalent to the asymptotic type III distribution, and is popular to the analysis of annual series of minimum values Z. The pdf is defined as:

$$f_Z(z) = \left(\frac{\kappa}{\alpha}\right)\left(\frac{z-b}{\alpha}\right)^{\kappa-1} e^{\left[-\left(\frac{z-b}{\alpha}\right)^{\kappa}\right]}$$

where $z \geq b$, $\alpha > b$ and $\kappa > 0$. The cdf is:

$$F_Z(z) = 1 - e^{\left[-\left(\frac{z-b}{\alpha}\right)^{\kappa}\right]}$$

Some expressions are:

- Mean

$$\mu_Z = b + \alpha\Gamma\left(1 + \frac{1}{\kappa}\right)$$

- Variance

$$\sigma_Z^2 = \alpha^2\left[\Gamma\left(1 + \frac{2}{\kappa}\right) - \Gamma^2\left(1 + \frac{1}{\kappa}\right)\right]$$

- Asymmetry coefficient

$$\gamma_1 = \frac{\Gamma\left(1 + \frac{3}{\kappa}\right) - 3\Gamma\left(1 + \frac{1}{\kappa}\right)\Gamma\left(1 + \frac{2}{\kappa}\right) + 2\Gamma^3\left(1 + \frac{1}{\kappa}\right)}{\left[\Gamma\left(1 + \frac{2}{\kappa}\right) - \Gamma^2\left(1 + \frac{1}{\kappa}\right)\right]^{1.5}}$$

  where $\Gamma(.)$ is the Gamma function.

The inverse of the Weibull cdf is:

$$z(F) = b + \alpha\left[-\ln\left(1 - F\right)\right]^{1/\kappa}$$

Note that when $\kappa = 1$ this distribution becomes the exponential distributio. Other properties of the Weibull distribution are given in textbooks tables.

Fréchet distribution

The **Fréchet distribution** is equivalent to the asymptotic type II distribution. The pdf is defined as:

$$f_X(x) = \frac{\theta}{a}\left(\frac{a}{\theta}\right)^{\theta+1} e^{\left[-\left(\frac{a}{x}\right)^{\theta}\right]}$$

where $x > 0$, and the scale and shape parameters are $a > 0$ and $\theta > 0$, respectively. The cdf is:

$$F_X(x) = e^{\left[-\left(\frac{a}{x}\right)^{\theta}\right]}$$

Some expressions are:

- Mean

  $$\mu_X = a\Gamma\left(1 - \frac{1}{\theta}\right)$$

  valid when $\theta > 1$
- Variance

  $$\sigma_X^2 = a^2\left[\Gamma\left(1 - \frac{2}{\theta}\right) - \Gamma^2\left(1 - \frac{1}{\theta}\right)\right]$$

  $$V_X = \left[\frac{\Gamma(1 - 2/\theta)}{\Gamma^2(1 - 1/\theta)} - 1\right]^{0.5}$$

  where the functions for σ_X^2 and V_X are valid for $\theta > 2$.

The quantile for the Fréchet pdf is given:

$$x(F) = a(-\ln F)^{-\theta}$$

Regarding to the exponential positive shape for $\theta > 0$, $x(F)$ increases faster than the Gumbel distribution when F increase. This means that Fréchet distribution produces a frequency curve of larger magnitudes. These two distributions are related through the logarithm transform, because when X follow a Fréchet pdf with parameters a and θ, $Y = \ln(X)$ follow a Gumbel pdf with parameters $\alpha = 1/\theta$ and $b = \ln(a)$. Other properties of the Fréchet distribution are given in textbooks tables.

GEV distribution

The **Generalized Extreme Value (GEV) distribution** is used for extreme maximum values, usually meteoroligical data. The pdf is:

$$f_X(x) = \frac{1}{a}\left[1 - \frac{b}{a}(x-c)\right]^{(1-b)/b} e^{-\left[1-\frac{b}{a}(x-c)\right]^{1/b}}$$

and the cdf is:

$$F_X(x) = e^{-\left[1-\frac{b}{a}(x-c)\right]^{1/b}}$$

where $a > 0$, b and c are the parameters of scale, shape and position, respectively. If $b > 0$ the distribution is bounded above representing a type III distribution which is useful for the analysis of extreme mimimum events. In contrast, if $b < 0$ the distribution is bounded below equivalent to a type II distribution suitable for extreme maximum events. If $b = 0$, the GEV distribution is transformed to a Gumbel distribution.

The fist two moments are:

- Mean

$$\mu_X = c + \frac{a}{b}[1 - \Gamma(1+b)]$$

- Variance

$$\sigma_X^2 = \frac{a^2}{b^2}[\Gamma(1+2b) - \Gamma^2(1+b)]$$

where $\Gamma(r)$ is the Gamma function evaluated in r. Note that while the equation for μ_X is valid for $b > -1$, the equation for σ_X^2 is valid for $b > -0.5$. A quantile is computed as:

$$x(F) = c + \frac{a}{b}\left[1 - (-\ln F)^b\right]$$

Other properties of the GEV distribution are given in textbooks tables.

Pareto distribution

The **Pareto distribution** is commonly used to analysed variables with long, straight tails. This distribution was initially implemented to analysed the wealth distribution in society where a low percentage concentrate the largest wealth. The pdf classic Pareto distribution or Pareto I distribution is:

$$f_x(x) = a_I c_I^{a_I} x^{a_I - 1}$$

where $x > c_I$ and $a_I > 0$. c_I and a_I are the position and shape parameters, respectively. The cdf is:

$$F_X(x) = 1 - \left(\frac{c_I}{x}\right)^{a_I}$$

The fist two moments are:

- Mean

  $$\mu_X = \frac{c_I a_I}{a_I - 1}$$

  Note that this is valid when $a_I > 1$, and ∞ on the contrary.

- Variance

  $$\sigma_X^2 = \left(\frac{cI}{a_I - 1}\right)^2 \frac{a_I}{a_I - 2}$$

  when $a_I > 2$. If $1 < a_I \leq 2$, $\sigma_X^2 = \infty$, and not exist when $a_I \leq 1$.

Pareto distribution

The Pareto I distribution is embeded in a more general Pareto II distribution. The pdf of this distribution is:

$$f_X(x) = \begin{cases} \frac{1}{b}\left(1 - a\frac{x}{b}\right)^{\frac{1}{a}-1} & \text{for } a \neq 0 \\ \frac{1}{b} e^{\left(-\frac{x}{b}\right)} & \text{for } a = 0 \end{cases}$$

The cdf is:

$$F_X(x) = \begin{cases} 1 - \left(1 - a\frac{x}{b}\right)^{\frac{1}{a}} & \text{for } a \neq 0 \\ 1 - e^{\left(-\frac{x}{b}\right)} & \text{for } a = 0 \end{cases}$$

where b is the scale parameter. Including a third parameter c, the position parameter, the pdf of the Pareto III distribution is:

$$f_X(x) = \begin{cases} \frac{1}{b}\left(1 - a\frac{x-c}{b}\right)^{\frac{1}{a}-1} & \text{for } a \neq 0 \\ \frac{1}{b} e^{\left(-\frac{x-c}{b}\right)} & \text{for } a = 0 \end{cases}$$

The cdf is:

$$F_X(x) = \begin{cases} 1 - \left(1 - a\frac{x-c}{b}\right)^{\frac{1}{a}} & \text{for } a \neq 0 \\ 1 - e^{\left(-\frac{x-c}{b}\right)} & \text{for } a = 0 \end{cases}$$

Note that when $c = 0$, the Pareto III distribution becomes the Pareto II distribution. Also, when $a < 0$ and $c = 0$ becomes the Pareto I distribution with $a_I = 1/a$ and $c_I = -b/a$.

Frequency curves for minimum events

- The commonly implemented probability distributions to analysed annual series of minimum events are: Weibull, Gumbel, Lognormal, Pearson type III and GEV. In the case of the Gumbel distribution, which is a unbounded distribution, can cause problem because negative values of the analysed variable can arise that are not realistic as is the case of some hydrological variables.
- These distribution can be fitted to the annual series of minimum values $z_i, i = 1, \cdots, n$ using a data transformation, that consist in multiplying the data by -1, which means that $x_i = -z_i$. This means that low values becomes high values and vice-versa. Thus, the fit of the probability distribution is made based on a annual series of maximum values x_i. At the end, the quantiles estimated for the frequency curve must be multiplied by -1 to return to the domain of Z.

Fitting process

The following steps summarize the process of fitting a probability distribution to an annual series.

1. Establishment of the annual series after applying statistical tests to detect outliers, and verify independence, homogeneity and stationarity in the series. Recall, that the size of the series affect in the estimation of the frequency curve and in the associated uncertainty. The literature thus recommends that the number of values of the series must be 25-30.
2. Selection of the pdf $f_X(x)$
3. Selection of the methods for parameter estimation $\hat{\theta}$
4. Estimation of magnitudes of events for selected return periods T, which depends upon the cdf $F_X(x)$ and the estimated parameters $\hat{\theta}$. Note that the T values should be less than 3-4 times the length (number of years) of the series. For maximum events:
$$\hat{X}_T = F_X^{-1}\left[1 - \frac{1}{T}\right]$$
and for minimum events:
$$\hat{X}_T = F_X^{-1}\left[\frac{1}{T}\right]$$
5. Estimation of the confidence limits for X_T for a significance level α. This means, confidence limits for the frequency curve.
6. Check through Q-Q plots and plots of the frequency curve if the chosen distribution is compatible with the annual series. Also, the use of the moments can verify the correcness of the chosen distribution with respect to the annual series. For instance, when γ_1 is close to zero, the Normal distribution could be considered. On the contrary, other distribution could be more appropriate. Also, hypothesis tests can be applied such as the Chi-square, Kolmogorov-Smirnov and Anderson-Darling, to verify the goodness of fit.

Anderson-Darling test

The **Anderson-Darling test** to verify the goodness of fit of a distribution is more adequate for the analysis of extreme events, because the test give more weight to the distribution tails. To perform this test on the following distributions: Gumbel, Fréchet, Normal, Lognormal, GEV, Gamma, Pearson type III and Log-Pearson type III, the first requirement is that the parameters estimators must be asymptotically efficient, usually estimated using the maximum likelihood method. For the Fréchet, Lognormal and Log-Pearson type III the annual series must be transformed taking logarithms and then fitted the transformed series to the Gumbel (max), Normal and Pearson type III distributions, respectively. After adjusting the data to any of these distributions using the maximum likelihood method the test statistics A^2 is estimated using:

$$A^2 = -n - \frac{1}{n}\sum_{i=1}^{n}(2i-1)\left[\ln(F_0(x_i)) + \ln(1 - F_0(x_{n-i+1}))\right]$$

where x_i are sorted in ascending order and $F_0(x_i)$ is the cdf evaluated in x_i. This is followed by the estimation of the parameter ω:

$$\omega = 0.116\left(\frac{A^2 - \xi_n}{\beta_n}\right)^{1.1751\eta_n} + 0.0403 \quad \text{for } \xi_n \leq A^2$$

$$\omega = \left[0.116\left(\frac{0.2\xi_n}{\beta_n}\right)^{1.1751\eta_n} + 0.0403\right]\left[\frac{A^2 - 0.2\xi_n}{\beta_n}\right] \quad \text{for } \xi_n > A^2$$

where the parameters ξ_n, β_n and η_n are computed using the expressions in the following table for different distributions and according to the size n of the annual series.

Anderson-Darling test

Tabla 2.30: Parámetros para calcular ω

Distribución	ξ_n	β_n	η_n
Gumbel / Frechét	$0{,}169\left(1+\frac{0{,}1}{n}\right)$	$0{,}229\left(1-\frac{0{,}2}{n}\right)$	$1{,}141\left(1+\frac{0{,}5}{n}\right)$
Normal y Lognormal	$0{,}167\left(1+\frac{0{,}3}{n}\right)$	$0{,}229\left(1-\frac{0{,}2}{n}\right)$	$1{,}147\left(1+\frac{0{,}5}{n}\right)$
GEV	$0{,}147\left(1-0{,}13b+0{,}21b^2+0{,}09b^3\right) \times\left(1+\frac{0{,}9}{n}-\frac{0{,}2}{\sqrt{n}}\right)$	$0{,}189\left(1+0{,}2b+0{,}37b^2+0{,}17b^3\right) \times\left(1-\frac{1{,}8}{n}\right)$	$1{,}186\left(1-0{,}04b-0{,}04b^2-0{,}01b^3\right) \times\left(1-\frac{0{,}7}{n}+\frac{0{,}2}{\sqrt{n}}\right)$
Gamma	$0{,}145\left(1+0{,}17r^{-1}+0{,}33r^{-2}\right) \times\left(1+\frac{2{,}0}{n}-\frac{0{,}3}{\sqrt{n}}-\frac{0{,}4}{r\sqrt{n}}\right)$	$0{,}186\left(1+0{,}34r^{-1}+0{,}3r^{-2}\right) \times\left(1-\frac{0{,}5}{n}-\frac{0{,}3}{\sqrt{n}}+\frac{0{,}3}{r\sqrt{n}}\right)$	$1{,}194\left(1-0{,}04r^{-1}-0{,}12r^{-2}\right) \times\left(1-\frac{1{,}8}{n}+\frac{0{,}1}{\sqrt{n}}+\frac{0{,}5}{r\sqrt{n}}\right)$
Pearson Tipo III Log Pearson Tipo III	$0{,}145\left(1+0{,}17b^{-1}+0{,}33b^{-2}\right) \times\left(1+\frac{2{,}0}{n}-\frac{0{,}3}{\sqrt{n}}-\frac{0{,}4}{b\sqrt{n}}\right)$	$0{,}186\left(1+0{,}34b^{-1}+0{,}3b^{-2}\right) \times\left(1-\frac{0{,}5}{n}-\frac{0{,}3}{\sqrt{n}}+\frac{0{,}3}{b\sqrt{n}}\right)$	$1{,}194\left(1-0{,}04r^{-1}-0{,}12b^{-2}\right) \times\left(1-\frac{1{,}8}{n}+\frac{0{,}1}{\sqrt{n}}+\frac{0{,}5}{b\sqrt{n}}\right)$

Para GEV: si $b > 0{,}5$, hacer $b = 0{,}5$
Para Gamma o Log Pearson Tipo III: si r o $b < 2$, hacer r o $b = 2$
Basado en Laio (2004)

Anderson-Darling test

The Anderson-Darling test shows that if $\omega < \omega_*$, where $\omega_* = 0.743, 0.581, 0.461$ and0.347 for significant levels of $\alpha = 0.01, 0.025, 0.05$ and0.1, H_0 is accepted, which means that the annual series can be represented by the chosen distribution. Alternatively, with the value of ω, F is calculated as as:

$$F(\omega) = \frac{1}{\pi\sqrt{\omega}}\left[e^{\left(-\frac{1}{16\omega}\right)}K_{1/4}\left(\frac{1}{16\omega}\right)\right] + \frac{1}{\pi\sqrt{\omega}}\left[1.118e^{\left(-\frac{25}{16\omega}\right)}K_{1/4}\left(\frac{25}{16\omega}\right)\right] \text{ for } \omega < 1.2$$

$$F(\omega) = 1 \text{ for } \omega \geq 1.2$$

where $K_{1/4}[.]$ is the modified Bessel function of order $1/4$. If $F(\omega) < (1-\alpha)$, H_0 is accepted; the lesser $F(\omega)$ the more suitable is the distribution chosen.

Distribution fitting

Annual series of maximum streamflow at Banco station, Magdalena River

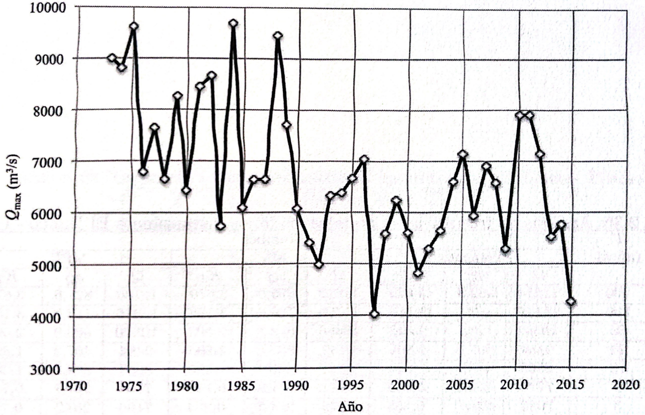


Figura 2.5: Serie de tiempo de la serie anual de caudales máximos instantáneos en El Banco

Distribution fitting

Annual series of maximum streamflow at Banco station, Magdalena River

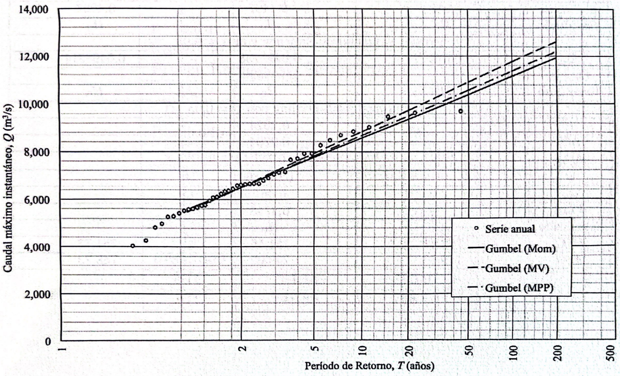


Figura 2.6: Curvas de frecuencia estimadas con Gumbel para la serie anual de caudales *máximos instantáneos* en El Banco

Annual series of maximum streamflow at Banco station, Magdalena River

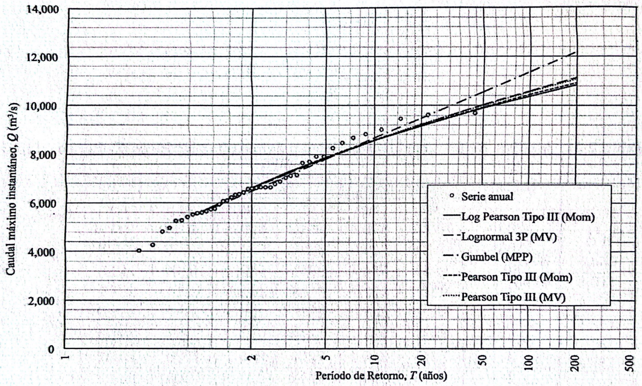


Figura 2.7: Curvas de frecuencia estimadas con Pearson y Log Pearson Tipo III y Lognormal 3P para la serie anual de caudales máximos instantáneos en El Banco

Distribution fitting

Annual series of maximum streamflow at Banco station, Magdalena River

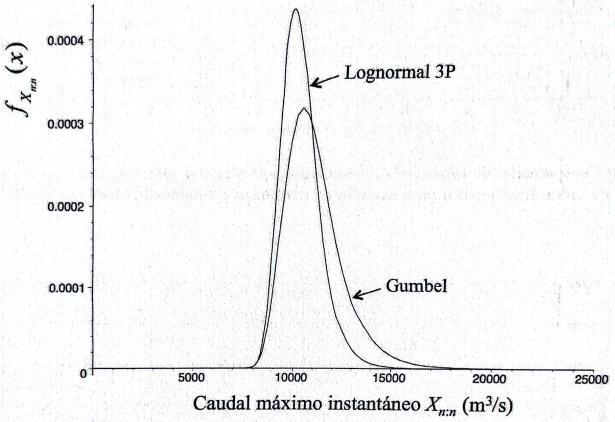


Figura 2.15: Funciones de densidad de probabilidad $f_{X_{n:n}}(x)$ del máximo caudal en 43 años en El Banco, a partir de las curvas de frecuencia ajustadas con Gumbel (MPP) y Lognormal 3P (MV)

Distribution selection

Generalities

- The distribution fitting processes described above can be applied to different types of distributions using different parameter-estimation methods.
- According to the literature, there is not one 'perfect' distribution for a specific study case, so multiple pdfs must be tried and initially analysed based on plotting and hypothesis testing.
- There are two types of procedures to select probability distribution: 1) Goodness-of-fit tests (e.g. Chi-squared and Kolmogorov-Smirnov goodness-of-fit tests), 2) information theory procedures (e.g. Akaike information criteria (AIC) and Bayesian information criteria (BIC)).

Distribution selection

AIC criteria

The **AIC** criteria is defined as

$$AIC = -2\ln\left[\prod_{i=1}^{n} f_X(x_i, \hat{\theta})\right] + 2p$$

where p is the number of parameters of $f_X(x)$, $\hat{\theta}$ represents the fitted values of the parameters, n is the size of the annual series and x_i is each value of the series. The first term of the right hand side of the equation represent the likelihood function and it is a measure of the inaccuracy of the estimation, while the second term account for the uncertainty in the estimation due to the number of parameters considering the distribution parsimony effects. The lesser AIC the better $f_X(x)$. The equation above for AIC is better for annual time series with $n \geq 30$. When $n/p < 40$, a corrected AIC must be calculated as:

$$AIC = -2\ln\left[\prod_{i=1}^{n} f_X(x_i, \hat{\theta})\right] + 2p\left(\frac{n}{n-p-1}\right)$$

BIC criteria

The **BIC** criteria result from a Bayesian analysis, so it is different to the AIC criteria. The BIC criteria is defined as

$$BIC = -2\ln\left[\prod_{i=1}^{n} f_X(x_i, \hat{\theta})\right] + \ln(n)p$$

The lesser BIC the better $f_X(x)$. The BIC criteria is more adequate for parsimonious distributions than the AIC criteria when $n \geq 8$.

Distribution selection

Tail classes

Regarding that one of the objective of the frequency analysis of extreme events is to estimate quantiles associated to high return periods, the fitted distributions must guarantee that their tails are well represented (inferior tail for minimum values and superior tails for maximum values) because extreme events are associated to tails. According to the tail shape, a distribution can have **light tails** if the probability of extreme values decreases very quickly, and **heavy tails** if the probability of extreme values (far from the mean) is relatively high. This means that for a heavy tail distribution, the magnitude is more extreme for the same return period. So before fitting different distributions is necessary to make exploratory analysis to identify the type of tail.

Tail classes

The following methodologies serve to identify tails:

1. Analyse the histogram of the data sample, once the sample is cleared up of outliers, the extreme values and the Q-Q plot; this last plots magnitudes from the observed and the theoretical distribution for different quantiles. The Q-Q plot is made for the observed magnitude vs each theoretical distribution. Note that if the Q-Q is lineal, the theoretical distribution is correct. In contrast, if the Q-Q plot is not lineal, this means that the annual time series have lighter or heavier tail than the theoretical distribution.
2. The **mean excess function**
$$e_i = \frac{\sum_{j=1}^{n}(x_j - x_i) I_{(x_j > x_i)}}{\sum_{j=1}^{n} I_{(x_j > x_i)}}$$
where $I_{(x_j > x_i)} = 1$ if $x_j > x_i$ and 0 otherwise, with the data sorted in descending order. So, e_i describe the expected excess over x_i. The plot i vs e_i is the empirical mean excess function. If the slope in this plot is horizontal, the data sample follows a exponential distribution, if the slope is positive the data follow a heavy tail distribution and if the slope is negative the data is associated to a light tail distribution.
3. The **Hill's plot** is based on the following estimator
$$\xi_i = \frac{1}{i-1}\left[\sum_{j=1}^{i-1} \ln x_j\right] - \ln x_i$$
with the data sorted in descending order and valid for $i \geq 2$. The Hill's plot is i vs ξ_i. If the plot tends to a constant value different of 0 the distribution tail is heavy, if the plot tends to 0 the distribution tail tends to be light as in the exponential distribution.

Distribution selection

Classification of asymptotic tail behaviour for some pdfs

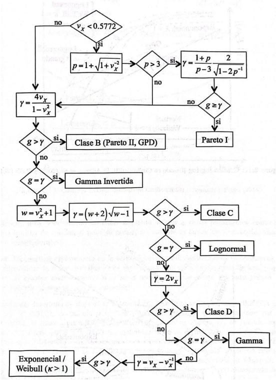
Clase	Tipo de cola	Características	Fdps	Comp. asintót.
A	Muy pesada	Cola de Pareto $0 < \alpha < 2$ Varianza infinita		$x \propto T^p$
B	Pareto	$F_X(x) = 1 - (u/x)^\alpha$ $\alpha > 0$ Cola decae con ley de potencia k momentos acotados ($k < \alpha$)	Pareto I Pareto II GPD	$x \propto T^p$
C	Regular-mente variable	Cola no es de Pareto pero es similar a Pareto, y cae asintótica-mente con ley de potencia	Fréchet Halphen inv. Tipo B Gamma inv. Log Pearson Tipo III Gamma Generalizada	$x \propto T^p$
Subclase Log-normal		Asintóticamente declina de forma similar a clases C o C/D	Lognormal 2P Lognormal 3P	$x \propto e^{(\ln T)^{0,5}}$
D	Subexpo-nencial	Cola decrece más lentamente que la Clase E	Gumbel Halphen Tipo A Pearson Tipo III Gamma Halphen Tipo B	$x \propto (\ln T)^p$
E	Exp.	fdps con $E[e^{sX}] = \infty$	Exponencial	$x \propto (\ln T)^p$
Normal		Cola más liviana que la Clase E	Normal	

Basado en El Adlouni et al. (2008) y Papalexiou et al. (2013)

Procedure to select the distribution

The following is a procedure to select the most adequate distributions for the annual series

1. The flow diagram below is based in the theoretical behaviour of the first three moments of the distributions, where v_X and g are the coefficients of variation and asymmetry of the annual series, respectively. Based on these values is possible to identify the distribution that fit the best to the annual series.



2. Some discriminatory plots based on the annual series:

- **log-log plot**: Plot of the $\log x_i$ vs $\log p_i$, where p_i the exceedance probability. If the plot is lineal, the distribution is C class; if the plot is concave is D or E class.
- **lognormality plot**: if $10v_X - 1.4 > g$ is suggested to plot the cdf of the annual serie with the normal probability in the horizontal axis and $\log x_i$ in the vertical axis. If the plot is lineal, the best distribution is lognormal and hypothesis testing of lognormality are recommended.
- **mean excess function**: If the slope tends to zero, the distribution is E class, if the slope is positive is D class. In this plot is recommended to plot those values greater than the median.

Distribution selection

Procedure to select the distribution

- **Hill's plot**: If the curve decrease asymptotically toward 0, the distribution is D or E class. However, if the plot converges to a different value, the distribution is C class.
- **Zenga plot**: This plot is useful to distinguish between lognormal and pareto distributions, where the n values of the annual series are sorted in ascending order. The x_i values are also grouped according to s values, so for instance, for x_j, $x_1 < \cdots < x_j < \cdots < x_s$ with frequencies n_j (number of times x_j is repeated in the annual series). So for $j = 1, \cdots, s$, the following variables are defined:

$$N_j = \sum_{i=1}^{j} n_j \qquad u_j = \frac{N_j}{n} \qquad T_j = \sum_{i=1}^{j} x_i n_i \qquad T = \sum_{j=1}^{s} x_j n_j$$

$$Q_j^- = \frac{T_j}{N_j} \qquad Q_j^+ = \begin{cases} \frac{T-T_j}{n-N_j} & j = 1, 2, \cdots, s-1 \\ x_s & j = s \end{cases} \qquad z_j = 1 - \frac{Q_j^-}{Q_j^+}$$

  The Zenga plot is u_j vs z_j. If the slope is positive, the distribution could be B or C class, if the slope tends to be horizontal it can be lognormal class, and if it negative the distribution is D or E class.

Note that for frequency analysis of minimum values in the lognormality plot must be analysed the inferior tail (lefthand side tail), while for the log-log, the Hill, the mean excess function and the Zenga plots must be made making a reflection transformation of the data. The reflection transformation is made by doing $x_i = -x_i$.

Distribution selection

Procedure to select the distribution

After the transformation, for the frequency analysis of minimum values in the log-log plot the horizontal axis correspond to $-\ln(-x_i)$ and the vertical axis correspond to $\ln[1 - F_{-X}(-x_i)]$. An for the Zenga plot $z_j = \frac{Q_j^-}{Q_j^x}$.

Once the most appropriate pdfs are identified, they are fitted to the data sample using the methods describe before and assessing the fitting using the AIC and BIC criteria. If there are more than one pdf, using the BIC value for each distribution, it is possible to estimate a weighted x_T value for different return periods (T) from the frequency curve. Based on the BIC values, posterior probabilities (b_j) are estimated to compute the weights (w_j). The weighted x_T are calculated as

$$w_j = \frac{b_j}{\sum_{i=1}^{m} b_i} \qquad b_j = e^{-0.5BIC_j} \qquad x_T = \sum_{j=1}^{m} w_j x_{T_j}$$

where m is the number of selected pdfs.

This procedure can be applied also to parameters such the s_X or the confidence limits to obtain a weighted version of them.